Why impact also needs energy

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Learning target: Explain why momentum does not tell the whole impact story.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 12; 3 course days later.*
Both become 3 times as large; p=mv and average force magnitude=change in momentum/time.

02. Explain a worked example (worked)
Equal momentum arrows; second energy bar twice the first.

03. Look and explain (model)
Second momentum doubles; kinetic energy quadruples.

04. Build the solution (practice)
4 J and 16 J.

05. Use the idea (reason)
Thermal energy, deformation, sound, motion of fragments; total energy conserved.

06. Choose a method (mixed) *Review: Unit 3, Day 3; 12 course days later.*
J=72 N·s right; v_f=18 m/s right.

07. Exit check (exit)
Factor 9.

08. STRETCH: Apply it another way (stretch)
Momentum doubles; kinetic energy quadruples. A valid numerical pair supports both.

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01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 12; 3 course days later.*
A has 1/3 of B's speed and kinetic energy: $v=p/m$, $K=p^2/(2\ m)$.

02. Explain a worked example (worked)
Equal momentum arrows; second energy bar twice the first.

03. Represent and interpret (model)
Second momentum doubles; kinetic energy quadruples.

04. Solve and justify (practice)
Speeds 6 and 2 m/s; energies 36 and 12 J.

05. Analyze the reasoning (reason)
Thermal energy, deformation, sound, motion of fragments; total energy conserved.

06. Choose a method (mixed) *Review: Unit 3, Day 3; 12 course days later.*
$J=12$ N·s; $\Delta v=15$ m/s in the force direction.

07. Exit check (exit)
One third.

08. STRETCH: Apply it another way (stretch)
From $v=p/m$, $K=p^2/(2\ m)$. Fourfold mass gives one quarter the energy at fixed momentum.